



Mental Health and Crime Rate

*Understanding the Relationship Using
Socioeconomic Factors, Data Exploration, and Clustering Analysis*

DSCI 510 | Final Project Presentation

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Introduction



WHAT

This project will explore the relationship between **mental health** status and **crime rates** in the United States, using potential socioeconomic factors at the **state-level**.

WHY

- useful to many **policymakers and think tanks**
- mental health affects **long-term relationships with people and behavior**
→ can relate to substance abuse, untreated disorders, impulsivity, etc.
- explores whether investing funds and resources into mental health programs, shelters, and aid would improve the quality of life for residents through the reduced crime
- early **preventative** measures for crime rates

HOW

- use publicly available data from the CDC, Bureau of Justice Statistics, and the US Census Bureau
- analyze correlation coefficients, variables, clustering patterns
- make conclusions regarding relationship

Data Sources



Source	Name/Description	Source URL/ Data URL	Data Type	Total Num of Data Points	Num Data Points (post-processing)	Format	Important Features
CDC (Center for Disease Control and Prevention)	~ Behavioral Risk Factor Surveillance System (BRFSS) - Mental Health Indicators ~ Mental health survey data in the United States	~ Source: https://data.cdc.gov/Mental-Health/Behavioral-Risk-Factor-Surveillance-System-BRFSS-M/5eh7-pjx8/about_data ~ Data: https://data.cdc.gov/resource/5eh7-pjx8.json?\$limit=50000	API	1086	54	JSON	area, area_name, question, percent
Bureau of Justice Statistics	~Crime Known to Law Enforcement 2024 ~ Estimates and RMSE for rate of violent victimization by state 2024	~ Source: https://bjs.ojp.gov/library/publications/crime-known-law-enforcement-2024 ~ Data: https://gist.githubusercontent.com/lrli1/d560e4d04b6b6e8315900bbe3224fc0c/raw/9dce7b56a93cc5e03e763b87e2dd5e56053525e0/Crime_Data.csv	Web CSV	53	45	CSV	state, rate per 100,000 estimate
United States Census Bureau	~ ACS 5-year Estimates (2020-2025), Selected Economic Characteristics ~ Economic demographic data for states in the US	~ Source: https://data.census.gov/table/ACSDP5Y.SPT2021.DP03?q=010XX00US\$3100000&d=ACS+5-Year+Estimates+Selected+Population+Data+Profiles ~ Data: https://gist.githubusercontent.com/lrli1/820584669eb70989d4d8d20b47d65e52/raw/809ccd245292102bcb9e9a8c2c51f537462376f8/gistfile1.txt	Web CSV	53	52	CSV	NAME, Percent!!PERCENTAGE OF FAMILIES AND PEOPLE WHOSE INCOME IN THE PAST 12 MONTHS IS BELOW THE POVERTY LEVEL!!All people, Percent!!HEALTH INSURANCE COVERAGE!!Civilian noninstitutionalized population!!No health insurance coverage, Percent!!EMPLOYMENT STATUS!!Population 16 years and over!!In labor force!!Civilian labor force!!Unemployed

Data Features



Dataset 1: CDC

- Percentage is standardized for
Question = 'Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?'

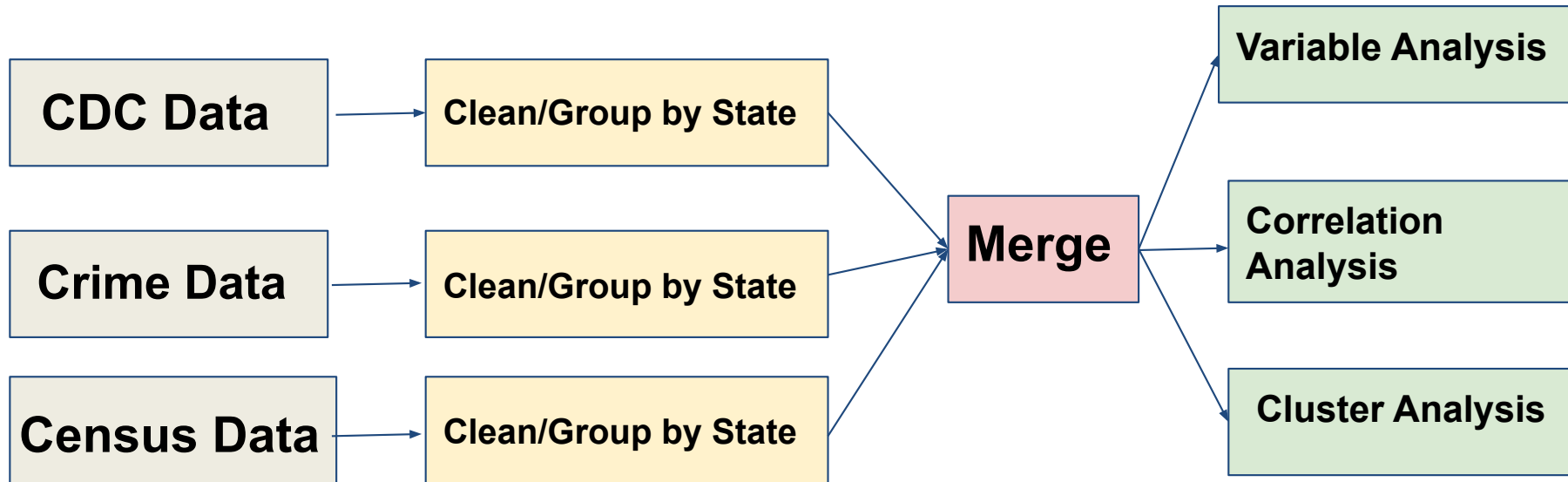
Dataset 2: Bureau of Justice Statistics

- **rate per 100,000 estimate** represents the number of violent crimes (rape, robbery, aggravated assault, homicide) standardized per 100,000 people in the population

Dataset 3: ACS

- % of families whose income in the past 12 months is below **poverty level**
- % of population with **no health insurance coverage**
- % of **unemployment status** for populations 16 and over in labor force

High-Level Workflow



Analysis



Part 1: Variable Analysis

- understand patterns for all individual variables using bar charts

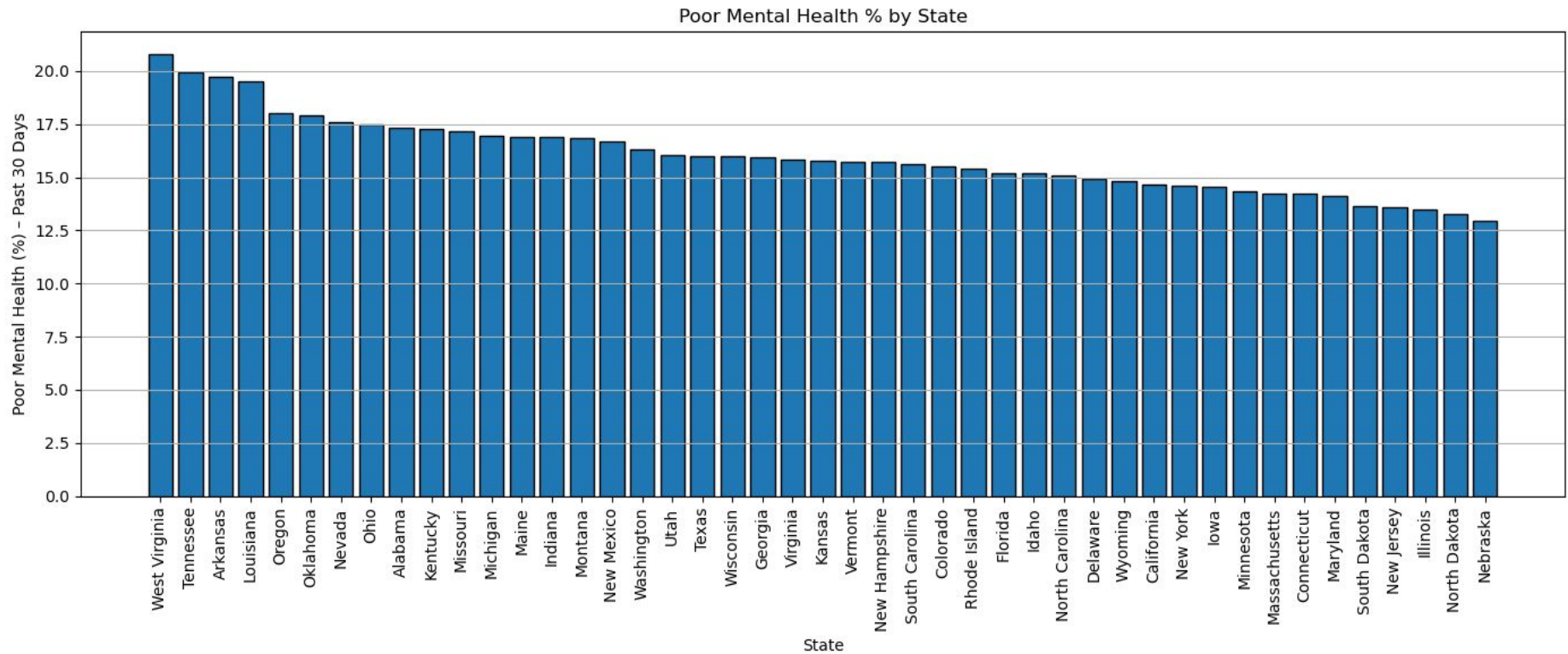
Part 2: Correlation Analysis

- Understand relationships between all variables using correlation coefficient & heat map
- Visualize relationship between mental health and crime using scatterplot

Part 3: Cluster Analysis

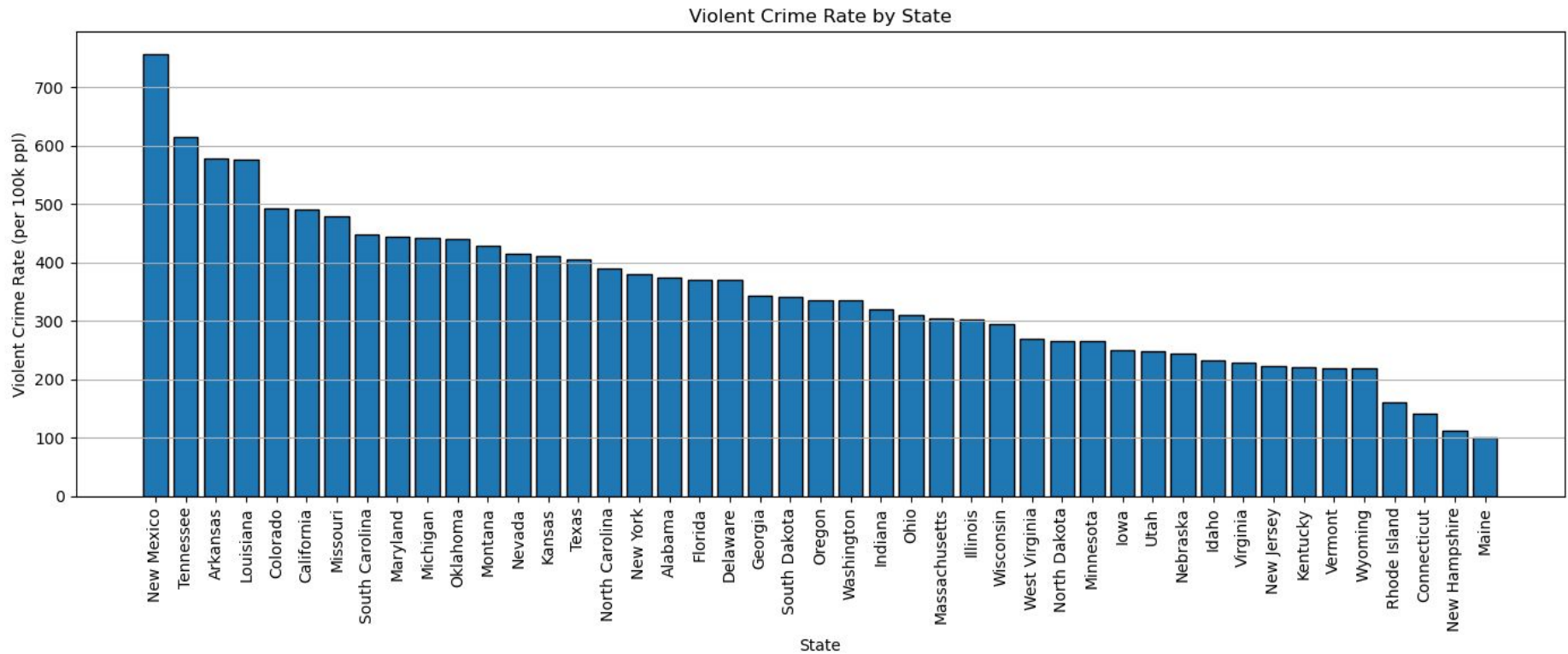
- Visualize how the data clusters without socioeconomic variables
- Visualize how the data clusters with socioeconomic variables
- find similarities between states in same cluster

Results



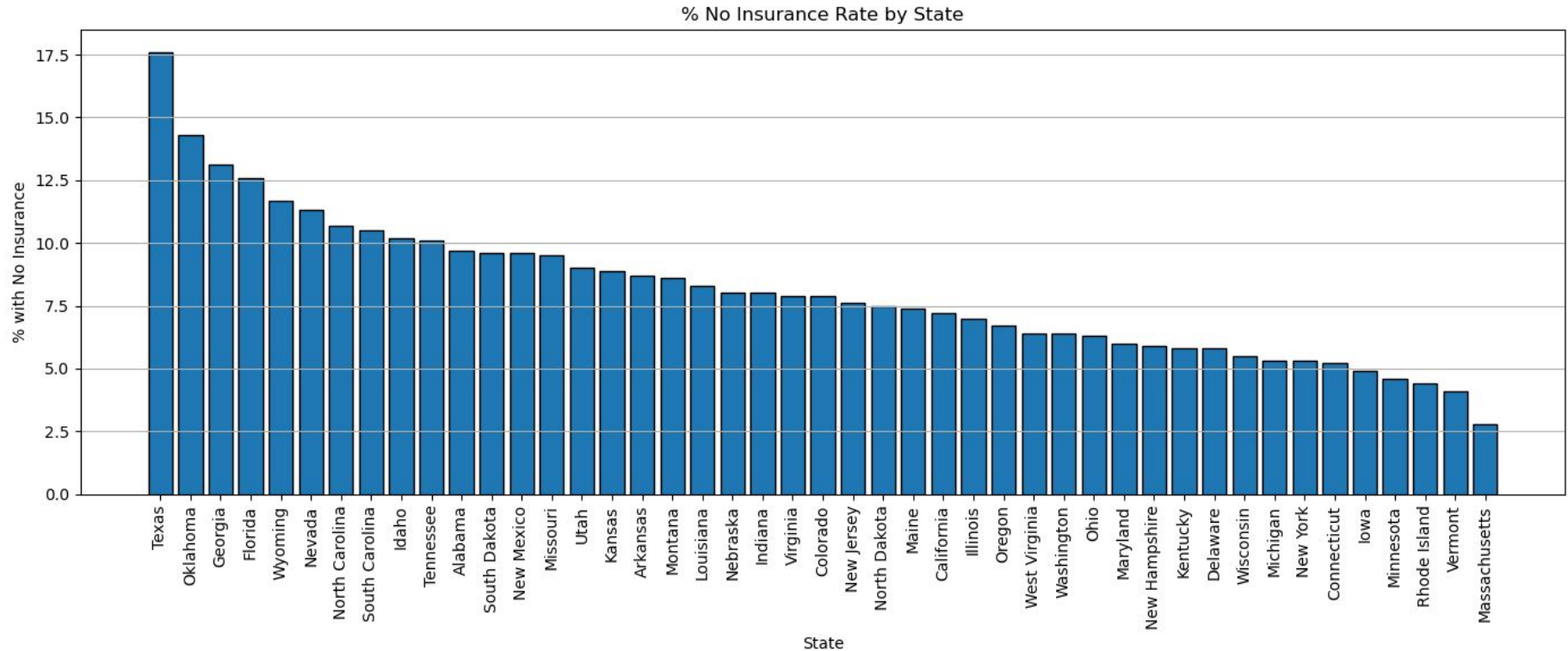
- West Virginia has highest rate of poor mental health in 30-day window
- Nebraska has lowest rate of poor mental health in 30-day window
- Roughly similar

Results



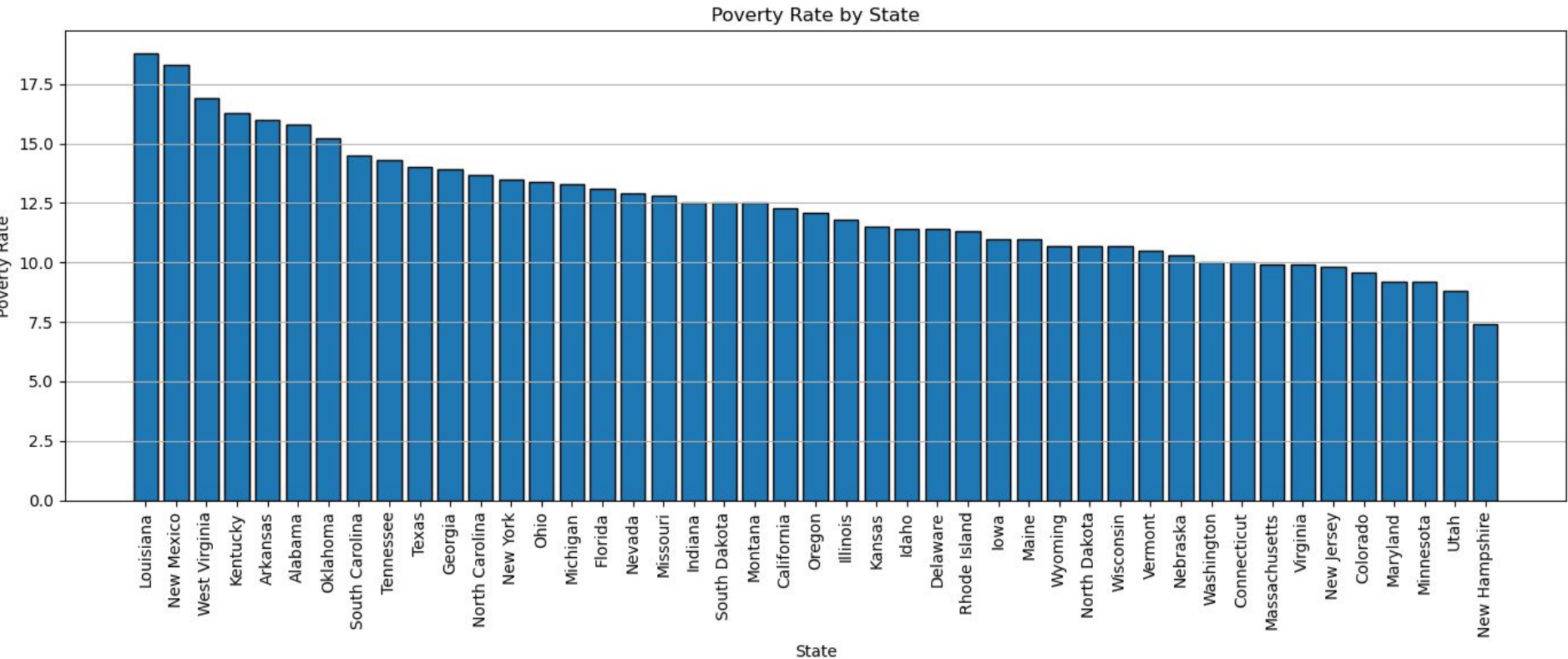
- New Mexico has highest crime rate per 100k people
- Maine has lowest crime rate per 100k people

Results



- Texas has the highest rate of people with no health insurance coverage
- Massachusetts has lowest rate of people with no health insurance coverage

Results

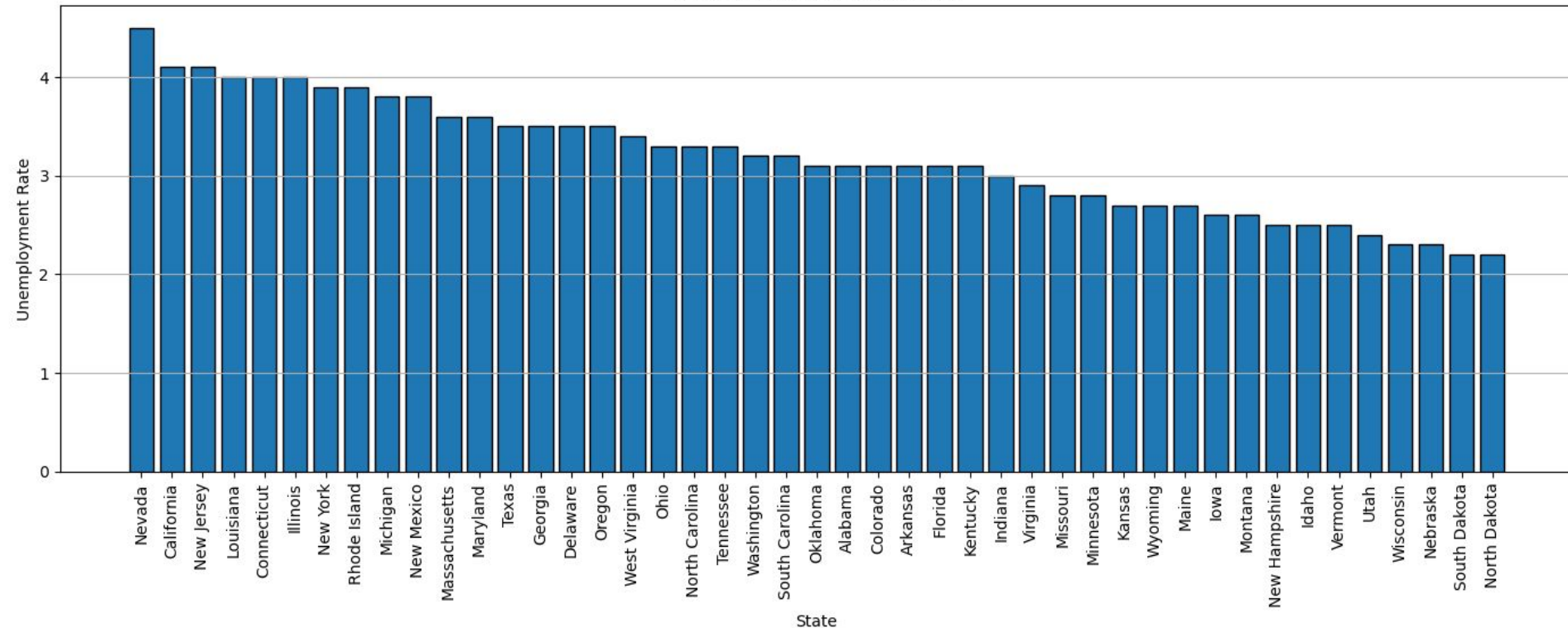


- Louisiana has the highest poverty rate
- New Hampshire has lowest poverty rate

Results



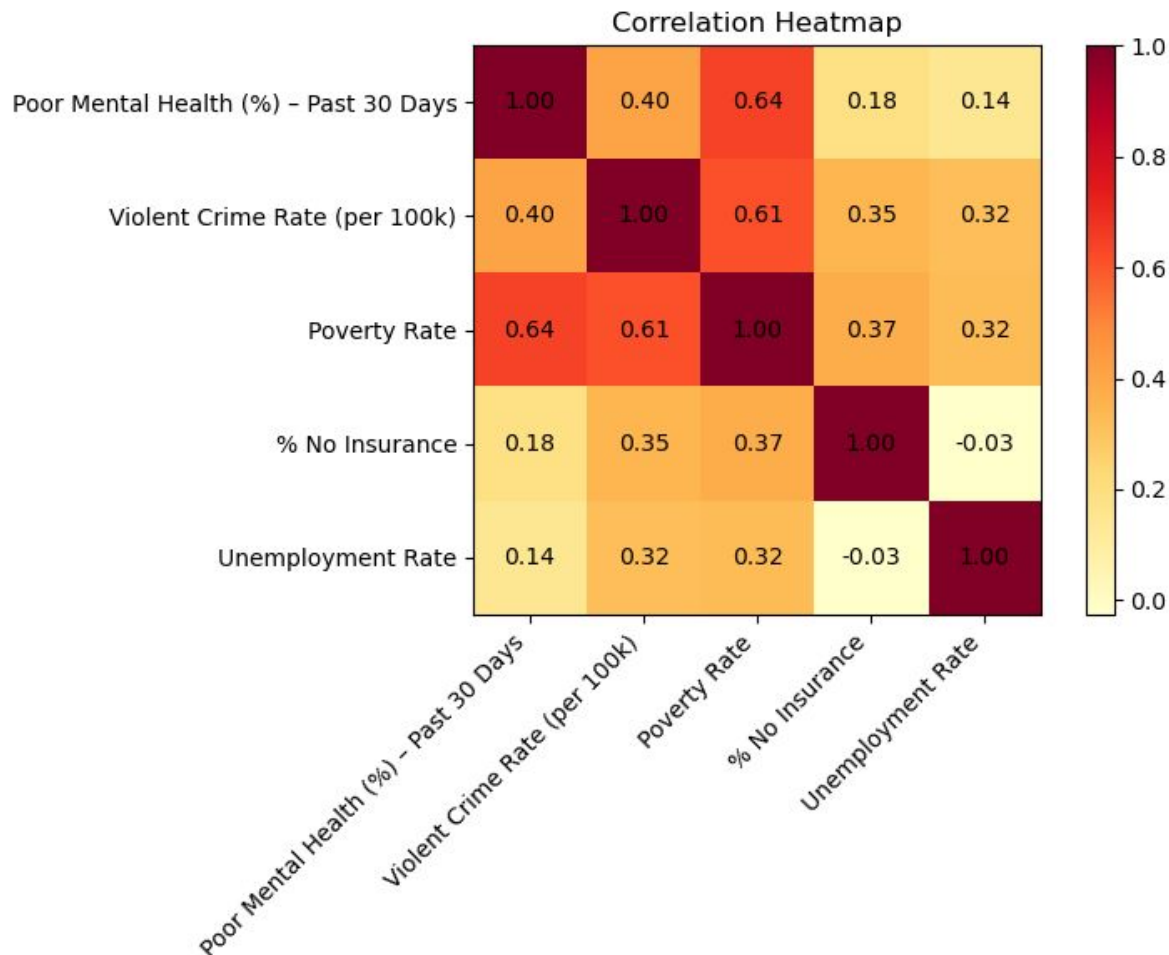
Unemployment Rate by State



- Nevada has the highest unemployment rate
- North Dakota has lowest unemployment rate



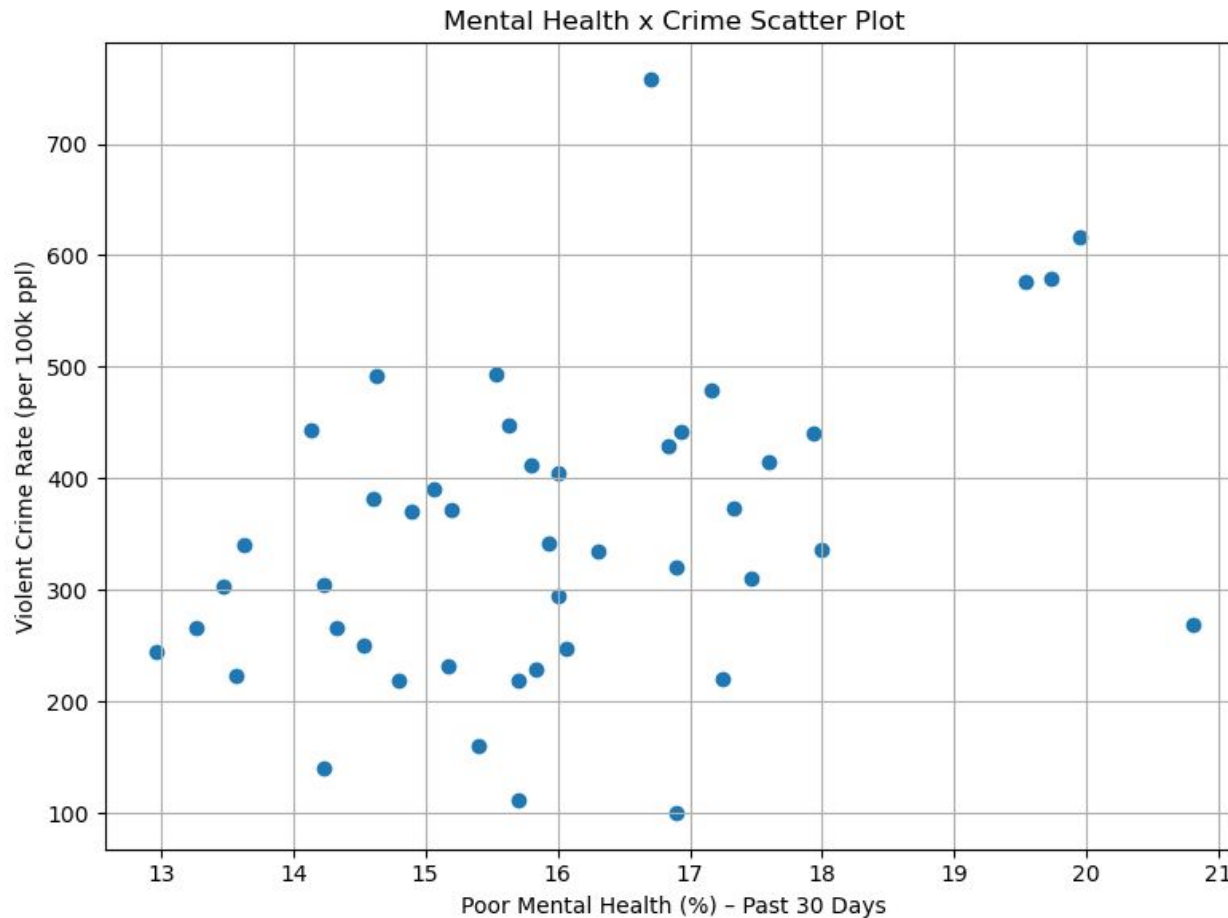
Results



- Violent crime and poor mental health have a correlation coefficient of **0.4**
- **Poverty rate** has the highest correlation with poor mental health
- **Poverty rate** also has the highest correlation with violent crime

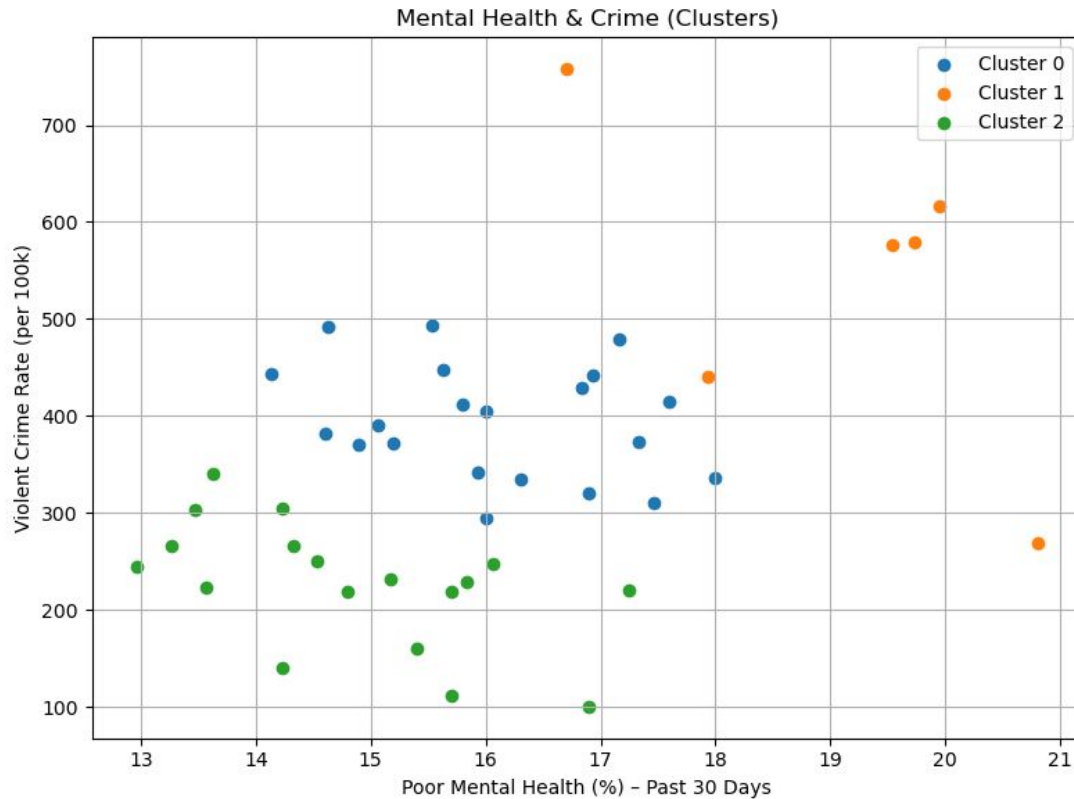
Poverty rate is an important → associated with both poorer mental health outcomes and elevated violent crime rates

Results



- Violent crime and poor mental health have a correlation coefficient of **0.4**
- Mental health and violent crime are **slightly positively correlated**, but not very strongly
- outliers and separate clusters exist

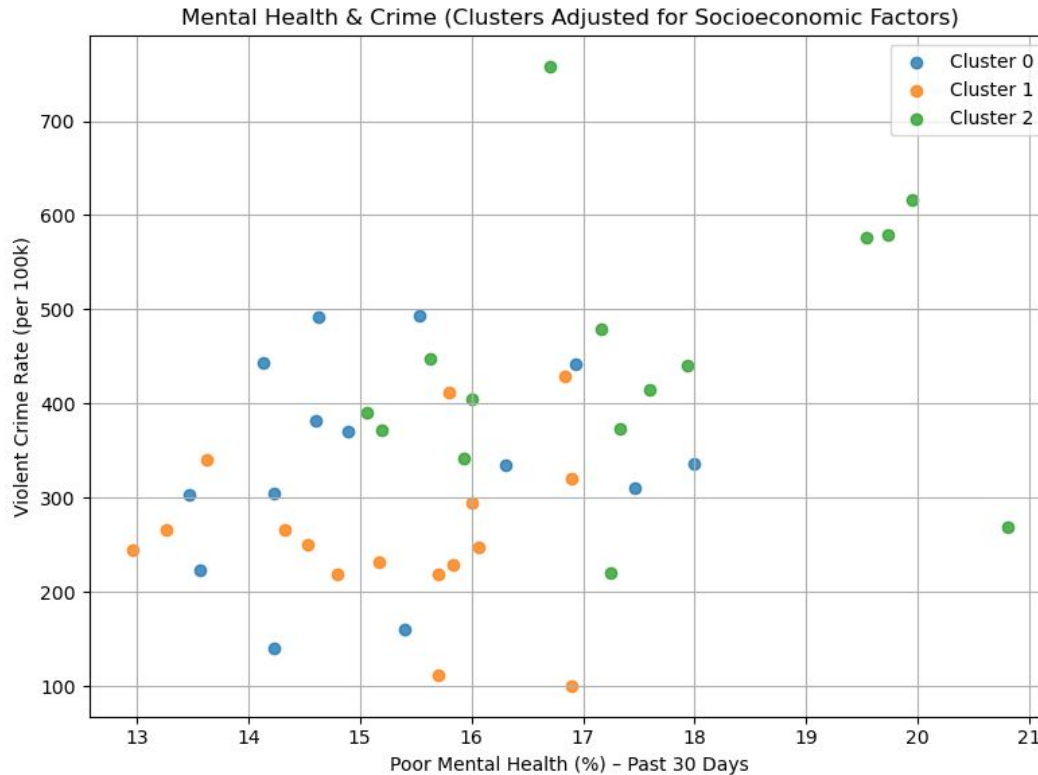
Results



- **Cluster 0:** Alabama, California, Colorado, Delaware, Florida, Georgia, Indiana, Kansas, Maryland, Michigan, Missouri, Montana, North Carolina, Nevada, New York, Ohio, Oregon, South Carolina, Texas, Washington, Wisconsin
- **Cluster 1:** Arkansas, Louisiana, New Mexico, Oklahoma, Tennessee, West Virginia
- **Cluster 2:** Connecticut, Iowa, Idaho, Illinois, Kentucky, Massachusetts, Maine, Minnesota, North Dakota, Nebraska, New Hampshire, New Jersey, Rhode Island, South Dakota, Utah, Virginia, Vermont, Wyoming

Cluster 0 represents states with moderate crime and moderate poor mental health. Cluster 1 represents states with high crime and high rates of poor mental health. Cluster 2 represents states with low crime and low rates of poor mental health. Without socioeconomic variables, there is a slight positive associative pattern → crime rates and mental health are related in some way.

Results



- **Cluster 0:** California, Colorado, Connecticut, Delaware, Illinois, Massachusetts, Maryland, Michigan, New Jersey, New York, Ohio, Oregon, Rhode Island, Washington
- **Cluster 1:** Iowa, Idaho, Indiana, Kansas, Maine, Minnesota, Montana, North Dakota, Nebraska, New Hampshire, South Dakota, Utah, Virginia, Vermont, Wisconsin, Wyoming
- **Cluster 2:** Alabama, Arkansas, Florida, Georgia, Kentucky, Louisiana, Missouri, North Carolina, New Mexico, Nevada, Oklahoma, South Carolina, Tennessee, Texas, West Virginia

After adding in socioeconomic variables, the clusters become more intertwined, and the relationship between mental health and crime is less distinct. No clear separation boundary between clusters that we can see in the 2D plane. Therefore, socioeconomic factors also drive mental health and crime. Additionally, states in cluster 0 seem to be from the coast. States in cluster 1 seem more in the Midwest. States in cluster 2 seem to be from the South. Other factors that could potentially drive patterns include urban/rural, political party, population, and more.

Findings



- **#1:** The clustering analysis groups changed when socioeconomic factors were added. Therefore, socioeconomic factors drive mental health and crime. The relationship between mental health and crime is partially driven by underlying socioeconomic circumstances.
- **#2:** Poverty drives both violent crime and mental health
- **#3:** Clustering with socioeconomic status places states in general groups of coastal states, midwestern states, and southern states. Common factors within these groups include urban/rural, political party, population, and more. These common factors can be added as further socioeconomic variables
- **#4:** Mental health and crime are related, but they are also importantly driven by extraneous variables. If a causal experiment were to be done to isolate the effects of mental health alone, socioeconomic variables must be accounted for.

Challenges



- Cleaning data to prepare for merge
 - isolating variables
 - grouping rows into state
 - dropping NA values
 - dropping incorrect values
- Finding accessible datasets
 - complete data with all states
 - free, publicly available
- Clustering
 - adding socioeconomic variables into clustering
 - Pre-processing data for K-Means analysis

Limitations



- **#1: Missing Socioeconomic variables**
 - many socioeconomic variables that significantly influence both violent crime and mental health were not included
 - seen in the similarities between states clustered together in latter clustering analysis
 - **includes:** urban vs rural, population density, political party, race, gender, education, policing, climate etc.
- **#2: Missing Data**
 - Missing States: Alaska, Arizona, Hawaii, Mississippi, Pennsylvania
 - Missing territories: Washington DC, Puerto Rico
- **#3: Can't Tell Causal Relationships**
 - causal relationship needs to be understood in order to implement real policy
 - analysis only accounts for correlation/associations



Thank You!

USCDornsife

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